# DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ

DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ
DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI

Çağrı Şükrü ÖZDOĞAN

İzmir Institute of Technology
October 2026

# DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI

**A Thesis Submitted to**
**the Graduate School of Engineering and Sciences of**
**İzmir Institute of Technology**
**in Partial Fulfillment of the Requirements for the Degree of**

**DOCTOR OF PHILOSOPHY**

**in Electrical and Electronics Engineering**

**by**
**Çağrı Şükrü ÖZDOĞAN**

**October 2026**
**İZMİR**

We approve the thesis of Çağrı Şükrü ÖZDOĞAN

**Examining Committee Members:**

_____

**Prof. Dr. Committee MEMBER 1**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 2**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 3**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 4**
Electrical Engineering, Example University

_____

**Prof. Dr. Committee MEMBER 5**
Electrical Engineering, Example University

10 October 2026

_____

**Prof. Dr. İlknur ÇİÇEK**
Supervisor
Electrical and Electronics Engineering,
IZTECH

_____

**Assoc. Prof. Gökhan IŞIK**
Co-Supervisor
Electrical and Electronics Engineering,
IZTECH

_____

**Prof. Dr. Ayşe YILMAZ**
Head of the Department
Electrical and Electronics Engineering

_____

**Prof. Dr. Deniz KAYA**
Dean of the Graduate School of
Engineering and Sciences

# PREFACE

This thesis combines three related articles.

# ACKNOWLEDGMENTS

I thank my supervisor and colleagues for their support.

# ABSTRACT

DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# ÖZET

DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI ÜZERİNDEKİ ETKİSİNİN KARŞILAŞTIRMALI OLARAK İNCELENMESİ DİJİTAL OKURYAZARLIĞIN ÇEVRİMİÇİ ÖĞRENME ORTAMLARINDA ÖĞRETMEN ADAYLARININ ÖZ YETERLİK ALGILARI

Bu doğrulama örneğinde kablosuz sistemlerde sinyal işleme incelenmektedir. Ölçüm sonuçları kuramsal tahminlerle karşılaştırılmış; yöntem, bulgular ve sınırlılıklar açıklanmıştır. Türkçe karakterler: ÇĞİÖŞÜ çğıöşü.

# TABLE OF CONTENTS

PREFACE..... iv
ACKNOWLEDGMENTS..... v
ABSTRACT..... vi
ÖZET..... vii
SYMBOLS AND ABBREVIATIONS..... ix
CHAPTER 1 ..... 1
INTRODUCTION
CHAPTER 2..... 2
ARTICLE 1
CHAPTER 3..... 3
ARTICLE 2
CHAPTER 4..... 4
ARTICLE 3
CHAPTER 5 ..... 5
CONCLUSIONS AND RECOMMENDATIONS
REFERENCES..... 6
APPENDIX A..... 7

# SYMBOLS AND ABBREVIATIONS

SNR: Signal-to-noise ratio

# CHAPTER 1

# INTRODUCTION

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 2

# ARTICLE 1

Publication citation: Example article 1, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 3

# ARTICLE 2

Publication citation: Example article 2, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 4

# ARTICLE 3

Publication citation: Example article 3, journal, volume, pages, DOI (test data).

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# CHAPTER 5

# CONCLUSIONS AND RECOMMENDATIONS

This verification specimen studies reproducible signal processing in wireless systems. The model compares measured observations with theoretical predictions and discusses limitations. The proposed method improves robustness while preserving computational efficiency. These sentences are deliberately long enough to test line wrapping, justification and paragraph indentation across the editor, exported PDF and Microsoft Word.

# REFERENCES

Smith, A. Signal Processing Methods; Example Press: İzmir, 2026.

# APPENDIX A

Supplementary measurement protocol.

# VITA

Çağrı Şükrü ÖZDOĞAN studied electrical engineering and researches signal processing.